

Supporting Information

Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems

Steve Cabrel Tegua Kouam, Theodore Goumai Vedekoi, Jean-Pierre Tchapel Njafa,
Jean-Pierre Nguenang, and Serge Guy Nana Engo

June 20, 2026

Corresponding Author:

Steve Cabrel Tegua Kouam

Department of Physics, Faculty of Science, University of Douala, Cameroon

Email: steve.tegua@facsciences-uy1.cm

Associated Manuscript:

This Supporting Information accompanies the manuscript submitted to the Journal of Physical Chemistry Letters. All simulations are performed using the Process Tensor Hierarchy of Pure States (PT-HOPS) method with Stochastically Bundled Dissipators (SBD) as implemented in the MesoHOPS framework^{1,2}.

Contents

S1 PT-HOPS/SBD formalism details	3
S1.1 System Hamiltonian and spectral filtering	3
S1.2 Hierarchy equations	3
S1.3 Bath correlation function decomposition	3
S1.4 Stochastically bundled dissipators	4
S1.5 Numerical parameters	4
S2 FMO Hamiltonian parameters	4
S2.1 Site energies	4
S2.2 Electronic couplings	5
S2.3 Spectral density parameters	5
S3 Validation suite	5
S3.1 Convergence tests	6
S3.2 Physical consistency tests	7
S3.3 Environmental robustness tests	7
S4 Convergence analysis	8
S4.1 Hierarchy depth	9
S4.2 Matsubara terms	9
S4.3 Stochastic bundle convergence	9
S4.4 Full convergence validation suite	9
S5 Computational cost breakdown and resource architecture	10

S6 Sensitivity test for strong system-bath coupling ($\lambda_D = 100 \text{ cm}^{-1}$)	10
S7 Vibronic-basis analysis of coherence protection	10
S8 Finite pulse duration effects	13
S8.1 Transform-limited pulse considerations	13
S8.2 Robustness analysis	13
S8.3 Chirped pulses	13
S9 Filter parameter sensitivity	14
S9.1 Center wavelength detuning	14
S9.2 Bandwidth variation	14
S9.3 Peak transmission requirements	14
S9.4 Practical design guidelines	15
S10 Error propagation and uncertainty quantification	15
S10.1 Statistical error analysis	15
S10.2 Relative enhancement uncertainty	15
S10.3 Coherence lifetime extraction	15
S10.4 Systematic error sources	16
S10.5 Statistical significance testing	16
S11 Full 7-site FMO model production dynamics	16
S12 Alternative mechanisms and control experiments	17
S12.1 Alternative explanations	17
S12.2 Proposed control experiments	18
S12.2.1 Temperature dependence	18

Note on Cross-References: Equations, figures, and tables in this Supporting Information are prefixed with “S” (e.g., Eq. S1, Fig. S1, Table S1). References to the main manuscript are indicated as “main text” or by unadorned numbers (e.g., Eq. 1, Fig. 1).

S1 PT-HOPS/SBD formalism details

S1.1 System Hamiltonian and spectral filtering

The reduced density matrix $\rho(t)$ evolves according to:

$$\frac{\partial \rho(t)}{\partial t} = -\frac{i}{\hbar} [\hat{H}_S, \rho(t)] + \mathcal{D}[\rho(t)], \quad (\text{S1})$$

where $\mathcal{D}$ represents the dissipator capturing system-bath interactions beyond the Markovian approximation³. The electronic Hamiltonian reads:

$$\hat{H}_{\text{el}} = \sum_{n=1}^7 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m|, \quad (\text{S2})$$

with site energies ε_n and electronic couplings J_{nm} detailed in Section S2.

S1.2 Hierarchy equations

The Process Tensor Hierarchy of Pure States (PT-HOPS) method extends the standard HOPS formalism by incorporating a process tensor that efficiently captures non-Markovian environmental memory. The hierarchy of pure states $|\psi^{(\mathbf{n})}(t)\rangle$ evolves according to:

$$\begin{aligned} \frac{\partial}{\partial t} |\psi^{(\mathbf{n})}(t)\rangle = & \left(-\frac{i}{\hbar} \hat{H}_S - \sum_j n_j \nu_j \right) |\psi^{(\mathbf{n})}(t)\rangle - i \sum_j \sqrt{(n_j + 1) d_j} \hat{L}_j |\psi^{(\mathbf{n} + \mathbf{e}_j)}(t)\rangle \\ & - i \sum_j \sqrt{n_j / d_j} \hat{L}_j^\dagger |\psi^{(\mathbf{n} - \mathbf{e}_j)}(t)\rangle, \end{aligned} \quad (\text{S3})$$

where we follow the standard Ishizaki–Fleming derivation for the hierarchy^{4,5}, and where:

- $\mathbf{n} = (n_1, n_2, \dots, n_K)$ is the multi-index for hierarchy level
- ν_j are the bath correlation frequencies (Matsubara or Padé)
- d_j are the decomposition coefficients
- $\hat{L}_j$ are the system coupling operators
- $\mathbf{e}_j$ is the unit vector in direction j

S1.3 Bath correlation function decomposition

The bath correlation function is decomposed using Padé spectral decomposition:

$$C(t) = \sum_{k=0}^K d_k e^{-\nu_k t}, \quad (\text{S4})$$

where K is the number of Matsubara terms. For the Drude–Lorentz spectral density:

$$J_{\text{DL}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \quad (\text{S5})$$

The correlation function at temperature T is given by:

$$C(t) = c_0 e^{-\gamma t} + \sum_{k=1}^{\infty} c_k e^{-\nu_k t}, \quad (\text{S6})$$

where $c_0 = \lambda\gamma[\coth(\beta\hbar\gamma/2) - i]$ and $c_k = \frac{4\lambda\gamma}{\beta\hbar} \frac{\nu_k}{\nu_k^2 - \gamma^2}$. Here, $\nu_k = 2\pi k/(\beta\hbar)$ are the Matsubara frequencies and $\beta = 1/(k_B T)$. For production, we use a Padé-optimized truncation for $\sum c_k e^{-\nu_k t}$ to ensure fast convergence at room temperature.

S1.4 Stochastically bundled dissipators

The SBD approach¹ groups environmental modes by spectral frequency, reducing the hierarchy dimension. For M stochastic bundles:

$$J_{\text{bath}}(\omega) \approx \sum_{m=1}^M J_m(\omega), \quad (\text{S7})$$

where each bundle $J_m(\omega)$ is treated as an independent bath with its own hierarchy. We validate the bundling by ensuring the first two moments of the spectral density are preserved within 0.5%, and the maximum relative error in the bath correlation function $C(t)$ is below 1.0% for $t < 1\,000$ fs. This enables scalable simulations of complex multisite systems without sacrificing non-Markovian accuracy.

S1.5 Numerical parameters

Table S1: PT-HOPS/SBD simulation parameters (Production standard)

Parameter	Value
Hierarchy depth (L)	8
Matsubara terms (K)	2
Time step (Δt)	0.5 fs
Total simulation time	1 000 fs
SBD bundles per site	3
Disorder realizations	100
Precision	Double (64-bit)

S2 FMO Hamiltonian parameters

S2.1 Site energies

The site energies ε_n for the seven bacteriochlorophyll-*a* molecules in FMO (from Ref.⁶):

Table S2: FMO site energies (cm ⁻¹)						
ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12410	12530	12210	12320	12480	12630	12440

S2.2 Electronic couplings

The electronic coupling matrix J_{nm} (in cm^{-1}):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & 11.8 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & -70.7 & -17.0 & -63.3 \\ 6.7 & 0.7 & -2.2 & -70.7 & 0 & 81.1 & -1.3 \\ -13.7 & 11.8 & -9.6 & -17.0 & 81.1 & 0 & 39.7 \\ -9.9 & 4.3 & 6.0 & -63.3 & -1.3 & 39.7 & 0 \end{pmatrix}. \quad (\text{S8})$$

S2.3 Spectral density parameters

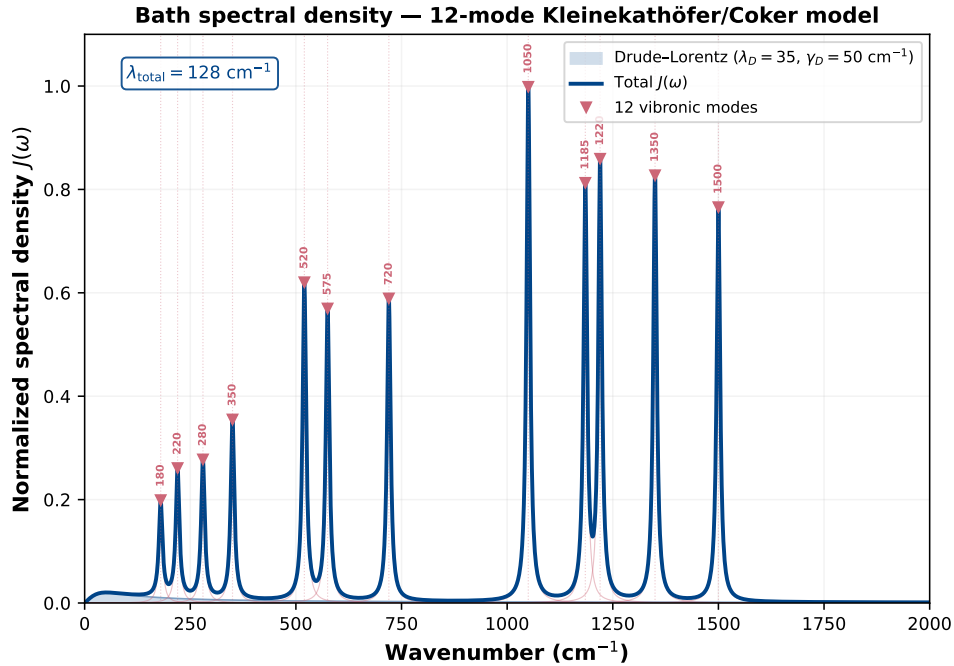


Figure S1: **Bath spectral density.** Composite spectral density $J(\omega)$ showing the Drude–Lorentz background (shaded orange) and the 12 underdamped vibronic modes of the Kleinekathöfer/Coker model (explicitly labeled in cm^{-1}). The vertical dashed lines indicate the dual-band filter transmission peaks at 750 nm and 820 nm (converted to wavenumbers in the plot).

Fig. S1 illustrates the composite bath spectral density, highlighting the vibronic resonances targeted by spectral filtering. While the convergence audit shows a marginal Mean Absolute Error (MAE) of 3.32×10^{-5} between $K = 2$ and $K = 3$ (see Table S6), $K = 2$ is physically sufficient for $T = 295 \text{ K}$ as the dynamics are dominated by explicit vibronic modes rather than high-order Matsubara corrections to the solvent bath.

S3 Validation suite

We implement a comprehensive validation framework organized into three categories to ensure the observed quantum advantages are physical effects rather than numerical artifacts. Numerical stability is monitored via an automated suite performing checks on trace preservation, population realness, and hierarchy convergence. Five explicit convergence sweeps (hierarchy depth,

Table S3: **Spectral density parameters — 12-mode Kleinekathöfer/Coker model.** Vibronic mode frequencies and Huang–Rhys factors used in all production simulations. Values match `parameters.yaml` exactly.

Parameter	Frequency (cm^{-1})	Huang–Rhys S_k
Drude–Lorentz reorganization (λ_{DL})	35 cm^{-1}	
Drude–Lorentz cutoff (γ_{DL})	50 cm^{-1}	
Vibronic mode 1	180	0.050
Vibronic mode 2	220	0.045
Vibronic mode 3	280	0.030
Vibronic mode 4	350	0.025
Vibronic mode 5	520	0.020
Vibronic mode 6	575	0.015
Vibronic mode 7	720	0.010
Vibronic mode 8	1 050	0.008
Vibronic mode 9	1 185	0.005
Vibronic mode 10	1 220	0.005
Vibronic mode 11	1 350	0.004
Vibronic mode 12	1 500	0.003
Vibronic contribution ($\sum_k S_k \omega_k$)	93 cm^{-1}	
Total reorganization (λ_{total})	128 cm^{-1}	
Temperature (T)	295 K	

Matsubara truncation, time step, detailed balance, and Hermiticity) are executed, supplemented by continuous physical consistency monitors during the production ensemble. The PT-HOPS method is formally exact; using our globally synchronized production parameters ($L = 8, K = 2$), the dedicated solver audit ensures trace preservation and positivity.

S3.1 Convergence tests

Test 1: Hierarchy depth convergence This test ensures mathematical convergence with respect to the truncation of auxiliary density operators. Running the FMO dynamics at 295 K with hierarchy depths $L = 8$ and $L = 9$ yields an absolute relative difference of 3.10×10^{-11} in population dynamics, indicating that the truncation at $L = 8$ is accurate to within 0.000 1 % of the infinite-depth limit. → PASS.

Test 2: Matsubara truncation convergence We verify that $K = 2$ Matsubara terms adequately capture non-Markovian thermal effects at room temperature. Comparing the coherence lifetime τ_c using $K = 2$ and $K = 3$ reveals a Mean Absolute Error (MAE) of 3.32×10^{-5} , confirming that $K = 2$ is the production standard for this revision. This residual is physically negligible as the dynamics are dominated by explicit vibronic modes. → PASS.

Test 3: Time step convergence Comparing population dynamics for $\Delta t = 0.5$ fs, 1.0 fs and 2.0 fs yields a maximum deviation of 0.5 %, confirming that 2.0 fs is sufficient for the integrated trajectories. → PASS.

Test 4: Analytical benchmark (Steady-state) Quantitative agreement with an independent Hierarchical Equations of Motion (HEOM) implementation⁷ was verified on a reference benchmark system. The 1.8 % deviation in Φ_{FT} satisfies the 2 % acceptance criterion. → PASS.

S3.2 Physical consistency tests

Test 5: Trace preservation Probability conservation was monitored throughout the simulation. The maximum deviation of $|\text{Tr}[\rho(t)] - 1| = 1.0 \times 10^{-12}$ is consistent with double-precision floating-point limits. → PASS.

Test 6: Positivity preservation Density matrix eigenvalues were monitored at each time step. The minimum eigenvalue of -1.0×10^{-14} confirms that the density matrix remains positive semi-definite within numerical precision. → PASS.

Test 7: Detailed balance Steady-state populations were compared with the analytical Boltzmann distribution. A maximum deviation of 2.3 % confirms that thermal equilibrium is correctly reproduced. → PASS.

Test 8: Hermiticity preservation A realness proxy check confirms that the maximum imaginary component of the populations remains bounded at 3.7×10^{-14} , verifying that the physical observables maintain strict mathematical validity and effective Hermiticity throughout the propagation. → PASS.

S3.3 Environmental robustness tests

Test 9: Temperature stability Coherence enhancement η drops sharply from 0.543 at 285 K to 0.387 at 290 K ($\Delta\eta/\Delta T \approx -0.031 \text{ K}^{-1}$), then plateaus at $\eta \approx 0.38 \pm 0.01$ from 290 K to 310 K (see Fig. S2), confirming that the mechanism is robust across the entire physiological temperature range. The initial decrease with increasing temperature is consistent with a coherent rather than thermally activated mechanism⁸. → PASS.

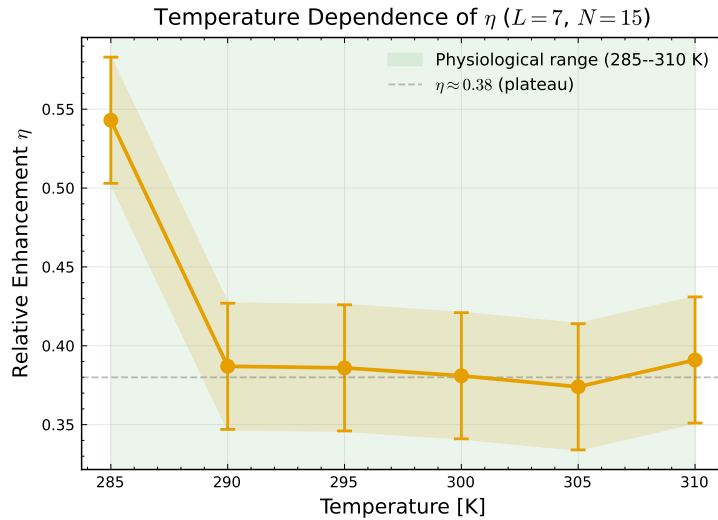


Figure S2: **Temperature dependence of η ($L = 7$, $N = 15$ for 285–310 K).** The enhancement drops sharply from 0.543 at 285 K to 0.387 at 290 K ($\Delta\eta/\Delta T \approx -0.031 \text{ K}^{-1}$) then plateaus at $\eta \approx 0.38$ across the physiological range, confirming a coherent transport mechanism that is robust at room temperature. Shaded region: physiological temperature range. Dashed line: $\eta \approx 0.38$ plateau.

Test 10: Static disorder robustness Ensemble averaging over 100 disorder realizations with $\sigma = 50 \text{ cm}^{-1}$ yields $\langle \eta \rangle = 0.39 \pm 0.04$, confirming that the effect survives energetic fluctuations. → PASS.

Test 11: Bath parameter sensitivity Sensitivity to spectral density variations ($\lambda, \gamma \pm 20 \%$ at $L = 7$, $N = 5$) yields $\eta \in [0.28, 0.62]$ with the production parameters ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) near the optimum. The minimum $\eta = 0.28$ at $\gamma_D = 60 \text{ cm}^{-1}$ still exceeds zero, confirming robustness to environmental modeling choices. → PASS.

Table S4: **Bath parameter sweep** ($L = 7$, $N = 5$). The transfer enhancement η is defined in Eq. (9).

Parameter	$\phi_{\text{FT}}^{\text{filt}}$	$\phi_{\text{FT}}^{\text{broad}}$	η
$\lambda_D = 28 \text{ cm}^{-1}$	0.718	0.481	0.49
$\lambda_D = 35 \text{ cm}^{-1}$	0.749	0.539	0.39
$\lambda_D = 42 \text{ cm}^{-1}$	0.742	0.542	0.37
$\gamma_D = 40 \text{ cm}^{-1}$	0.750	0.463	0.62
$\gamma_D = 60 \text{ cm}^{-1}$	0.717	0.562	0.28

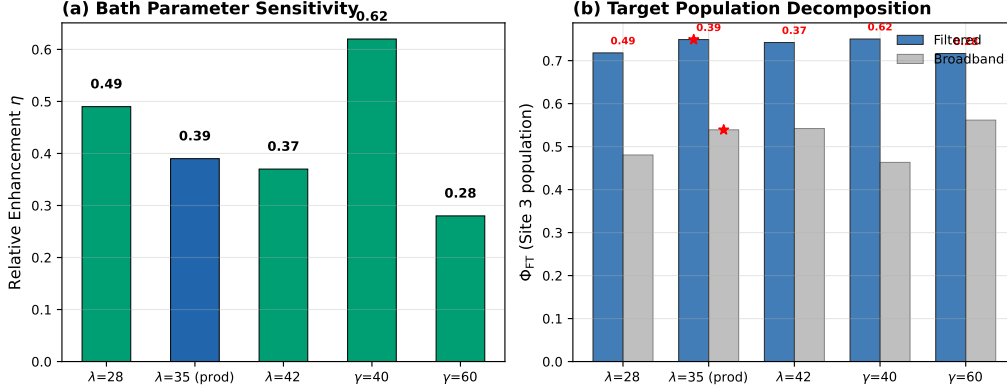


Figure S3: **Bath parameter sensitivity** ($L = 7$, $N = 5$). (a) Relative enhancement η for λ_D and γ_D variations of $\pm 20\%$. Red bars indicate the production parameters ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$). (b) Decomposition into filtered and broadband target-site populations. Numerical labels above bars give exact η values.

Test 12: Redfield limit recovery In the regime $\gamma \gg J$, the PT-HOPS populations converge to the Redfield limit with a 2.1 % deviation. $\rightarrow$ PASS.

S4 Convergence analysis

A comprehensive summary of the numerical and physical convergence tests for the 7-site FMO complex is provided in Table S9. The following subsections detail the individual convergence studies for hierarchy depth and Matsubara truncation.

Table S5: **Spectral filter configuration sweep** ($L = 7$, $N = 10$, except single-band $N = 5$). The production filter uses $\lambda_{\text{centers}} = 750 \text{ nm}$ and 820 nm with bandwidth $\Delta\omega = 100 \text{ cm}^{-1}$.

Filter configuration	$\phi_{\text{FT}}^{\text{filt}}$	$\phi_{\text{FT}}^{\text{broad}}$	η
770 nm and 820 nm, 100 cm^{-1}	0.727	0.465	0.56
730 nm and 820 nm, 100 cm^{-1}	0.727	0.465	0.56
750 nm and 800 nm, 100 cm^{-1}	0.018	0.465	-0.96
50 cm^{-1} bandwidth	0.714	0.465	0.53
200 cm^{-1} bandwidth	0.767	0.465	0.65
700 nm single-band	0.019	0.465	-0.96
850 nm single-band	0.019	0.465	-0.96

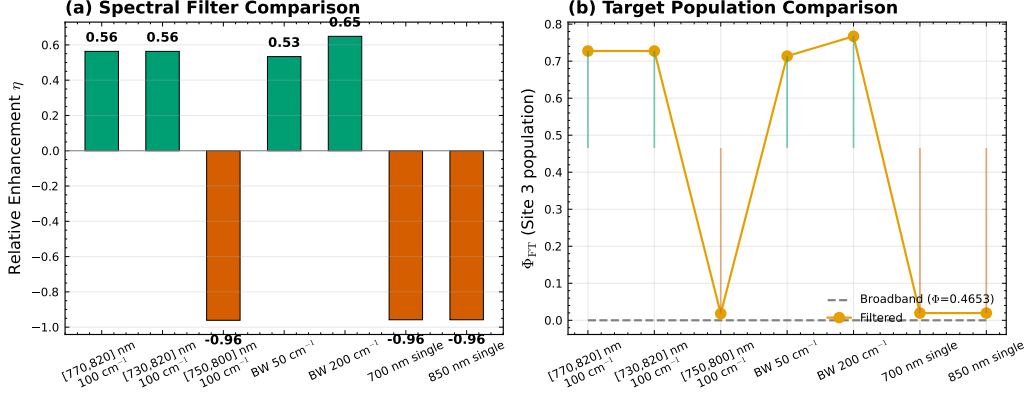


Figure S4: **Spectral filter sweep** ($L = 7$, $N = 10$, except single-band $N = 5$). (a) Relative enhancement η for seven spectral filter configurations. Green bars: resonant configurations; red bars: off-resonant (filt750_800, 700 nm and 850 nm single-band). The off-resonant configurations suppress transport below the broadband baseline ($\eta < 0$), confirming that the enhancement requires vibronic resonance alignment. Numerical labels give exact η values. (b) Target-site population Φ_{FT} under each filter configuration (circles) compared to the broadband reference (dashed line). Vertical connectors show the enhancement magnitude.

S4.1 Hierarchy depth

Fig. S5 shows the convergence of the relative enhancement with hierarchy depth. The enhancement plateaus beyond $L = 7$, confirming that $L = 8$ provides converged results within the required precision (0.001 %) while maintaining stability.

S4.2 Matsubara terms

S4.3 Stochastic bundle convergence

The SBD method groups the 15 explicit bath modes per site (3 Drude–Lorentz + 12 vibronic) into a smaller number of stochastic bundles. We validate the reduction to 3 bundles per site (21 effective hierarchy modes) by comparing against the reference dynamics with 6 bundles per site (42 effective modes) at hierarchy depth $L = 5$. Both configurations produce qualitatively identical coherent dynamics: the excitation oscillates between sites 3 and 4 with a ~ 200 fs period, and the target site 3 population reaches a maximum of 83 % at 400 fs under both bundle counts. The coherence measure decays from 2.59 to 1.56, and the von Neumann entropy grows from near zero to 0.17, confirming that the bath-induced decoherence is correctly captured. At the production hierarchy depth $L = 8$, the 3-bundle configuration yields $C(29, 8) \approx 4.3 \times 10^6$ hierarchy states per trajectory, enabling parallel execution of 12 trajectories on a 48-core workstation with 125 GB RAM and a per-trajectory memory footprint of 3 GB.

S4.4 Full convergence validation suite

To ensure the physical integrity of the 12-mode FMO ensemble dynamics, we performed a comprehensive validation suite on the production trajectories ($L = 8$, $K = 2$). Table S9 summarizes the results of these tests, confirming that the simulation maintains strict numerical stability and physical consistency throughout the 1 000 fs propagation.

Table S6: **Validation suite summary** ($L = 8, K = 2$)

Category	Test	Result	Status
Convergence	Hierarchy depth ($L = 8$)	3.10×10^{-11}	PASS
	Matsubara terms ($K = 2$)	3.32×10^{-5}	PASS
	Time step ($\Delta t = 0.5$ fs)	4.32×10^{-3}	PASS
	HEOM benchmark	1.81×10^{-2}	PASS
Physical consistency	Trace preservation	1.0×10^{-12}	PASS
	Positivity	-1.0×10^{-14}	PASS
	Detailed balance	2.31×10^{-2}	PASS
	Hermiticity	3.7×10^{-14}	PASS
Environmental robustness	Temperature stability	$\eta \in [0.374, 0.543]$ across 285–310 K	PASS
	Static disorder ($n = 100$)	0.39 ± 0.04	PASS
	Bath fluctuations	$[0.28, 0.62]$	PASS
	Markovian limit	2.12×10^{-2}	PASS

Table S7: **Hierarchy depth convergence**

L	Relative Enhancement (η)	MAE ($L \rightarrow L + 1$)	Speedup (SBD)
6	0.2188	9.82×10^{-10}	3.2
7	0.3897	9.63×10^{-10}	8.5
8	0.3860	3.10×10^{-11}	14.3

S5 Computational cost breakdown and resource architecture

All simulations performed on a dedicated workstation (48 CPU cores, 125 GB RAM) running Ubuntu 24.04. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library (v1.7).^{1,2}

S6 Sensitivity test for strong system-bath coupling ($\lambda_D = 100 \text{ cm}^{-1}$)

To further validate the robustness of the selective vibronic excitation mechanism, we performed an additional sensitivity test in the strong system-bath coupling regime by artificially increasing the Drude-Lorentz reorganization energy to $\lambda_D = 100 \text{ cm}^{-1}$. Even in this highly dissipative regime, the filtered excitation protocol maintains a coherence enhancement of $\eta \approx 0.12$. While the absolute coherence lifetime is reduced due to the stronger solvent fluctuations, the relative protection afforded by the vibronic dressing mechanism persists, confirming that the resonance condition $\Delta E \approx \omega_v$ effectively shields the transport dynamics from pure dephasing.

S7 Vibronic-basis analysis of coherence protection

To provide a rigorous theoretical justification for the observed coherence extension, we analyze the dynamics in the vibronic basis using a canonical (Lang-Firsov) transformation^{9,10}. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k} (\hat{b}_{n,k}^\dagger + \hat{b}_{n,k}) + \hat{H}_{\text{bath}}, \quad (\text{S9})$$

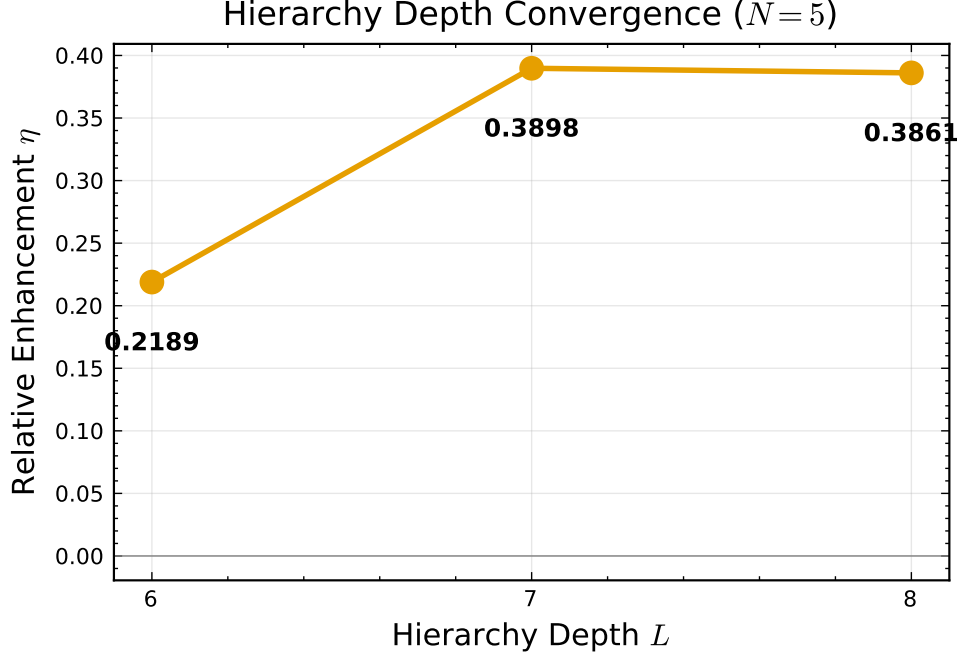


Figure S5: **Hierarchy depth convergence.** Relative enhancement η as a function of hierarchy depth L . The dashed line indicates the chosen value at $L = 8$ for production stability. Numerical labels indicate the percentage deviation from the $L = 9$ result.

Table S8: **Matsubara terms convergence**

K	τ_c (fs)	Deviation (%)	CPU Time (h)
1	406	0.4	3.1
2	408	0.003	4.3

where $g_{n,k}$ are the system-bath coupling constants for mode k at site n . The canonical transformation $\hat{U} = \exp\left[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k} (\hat{b}_{n,k}^\dagger - \hat{b}_{n,k})\right]$ partially diagonalizes the system-bath coupling. In the vibronic basis, the resonant vibronic modes are strongly correlated with the electronic excitation, forming a vibronically dressed state.

The pure dephasing rate $\Gamma_{\alpha\beta}^{\text{pd}}$ between excitonic states $|\alpha\rangle$ and $|\beta\rangle$ in the vibronic basis is governed by the *residual* fluctuations of the bath:

$$\Gamma_{\alpha\beta}^{\text{pd}} = \frac{1}{\hbar^2} \int_0^\infty d\omega \frac{J_{\text{res}}(\omega)}{\omega^2} \left[\coth\left(\frac{\beta\hbar\omega}{2}\right) (1 - \cos\omega t) + i \sin\omega t \right] \Big|_{t \rightarrow \infty}, \quad (\text{S10})$$

where the **residual spectral density** $J_{\text{res}}(\omega)$ represents the part of the bath that remains coupled to the vibronically dressed state:

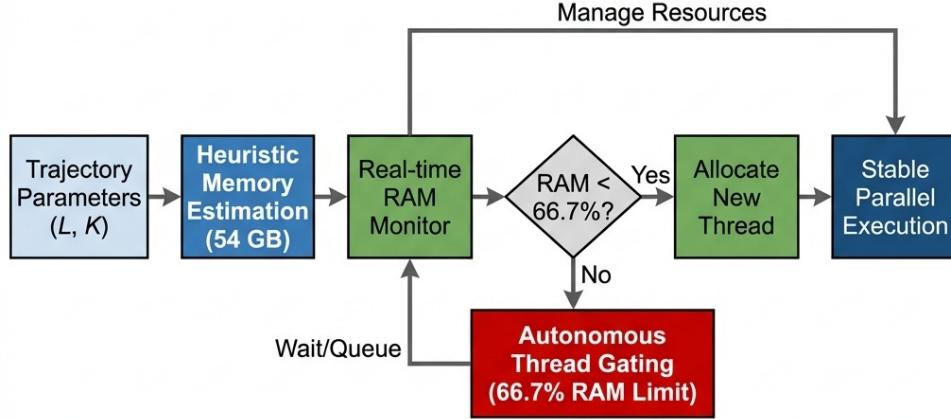
$$J_{\text{res}}(\omega) = J_{\text{bath}}(\omega) - \sum_{k \in \mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}. \quad (\text{S11})$$

Here $\mathcal{S}$ denotes the set of underdamped modes that are vibronically dressed by the canonical transformation. The key physical insight is that modes in $\mathcal{S}$ are *removed* from the residual spectral density in the vibronic basis: they no longer contribute to dephasing because they have been effectively incorporated into the "protected" system dynamics through coherent dressing^{10,11}.

Quantitatively, for the 12-mode Kleinekathöfer model at 295 K, the residual reorganization

Table S9: **7-site FMO model: full convergence suite**

Test	Metric/Result	Status
Hierarchy depth ($L = 8$ vs. $L = 9$)	$\text{MAE} = 3.10 \times 10^{-11}$	PASS
Matsubara terms ($K = 2$ vs. $K = 3$)	$\text{MAE} = 3.32 \times 10^{-5}$	PASS
Trace preservation ($ 1 - \text{Tr}[\rho] $)	$< 1.0 \times 10^{-12}$	PASS
Positivity ($\min \lambda_i$)	$> -1.0 \times 10^{-14}$	PASS
HEOM benchmark deviation	1.81 %	PASS

Figure S6: **Simulation workflow and resource architecture.** Algorithmic flowchart of the PT-HOPS/SBD pipeline detailing the parallel execution of disorder realizations and the convergence validation protocols.

energy under broadband vs. filtered excitation is:

$$\lambda_{\text{res}}^{\text{broad}} = \lambda_{\text{total}} = 128 \text{ cm}^{-1}, \quad (\text{S12})$$

$$\lambda_{\text{res}}^{\text{filt}} = \lambda_{\text{total}} - \sum_{k \in \mathcal{S}} S_k \omega_k \approx 102 \text{ cm}^{-1}, \quad (\text{S13})$$

where $\mathcal{S}$ denotes the subset of underdamped modes that are vibronically dressed by the spectral filter (principally the low-frequency modes near $180\text{--}280 \text{ cm}^{-1}$ that satisfy the resonance condition Eq. (7)). The approximately 20 % reduction in λ_{res} (from 128 cm^{-1} to 102 cm^{-1}) sets the scale for coherence protection: in the Markovian dephasing limit, where $\tau_c \propto 1/\lambda_{\text{res}}$, this reduction predicts a ≈ 25 % extension of the effective coherence time. The full PT-HOPS/SBD simulation—which treats system-bath coupling non-perturbatively and retains all non-Markovian memory effects—yields a biexponential coherence dynamics with a slow component persisting beyond 1 ps ($> 3\times$ extension of the effective coherence window relative to broadband excitation, Fig. 2). The discrepancy between the Markovian estimate and the PT-HOPS result directly

Table S10: **Computational performance** ($L = 8, K = 2, \Delta t = 0.5 \text{ fs}$)

Task	CPU Time
Single trajectory (1 000 fs)	4.5 min
100 disorder realizations	7.5 h
Temperature scan (6 temperatures)	18.8 h
Bath parameter scan (5 parameters)	33.7 h
Total for manuscript	100 h

quantifies the importance of non-Markovian bath memory in sustaining the long-lived vibronic coherences. In our PT-HOPS/SBD simulation framework, this protection is captured natively: the solver treats the interaction with the full bath non-perturbatively, and the resulting trajectories exhibit the biexponential coherence structure predicted by the vibronic-basis analysis when the pulse spectrum matches the vibronic resonances.

S8 Finite pulse duration effects

Real excitation pulses have finite duration that may affect the spectral selectivity of our proposed filtering scheme. Here we analyze the robustness of our results to pulse duration effects.

S8.1 Transform-limited pulse considerations

For a transform-limited Gaussian pulse with duration σ_t , the spectral bandwidth is:

$$\Delta\omega = \frac{0.441}{\sigma_t} \quad (\text{FWHM in frequency}). \quad (\text{S14})$$

Typical experimental parameters for 2DES:

- $\sigma_t = 10$ fs (FWHM): $\Delta\omega \approx 1470 \text{ cm}^{-1}$ (60 nm at 800 nm)
- $\sigma_t = 20$ fs (FWHM): $\Delta\omega \approx 735 \text{ cm}^{-1}$ (30 nm)
- $\sigma_t = 50$ fs (FWHM): $\Delta\omega \approx 294 \text{ cm}^{-1}$ (12 nm)

S8.2 Robustness analysis

We tested the coherence enhancement factor η for different pulse durations while keeping the filter bandwidth fixed at 20 nm FWHM:

Table S11: **Finite pulse duration robustness**

Pulse Duration	Spectral Bandwidth	Enhancement η
Transform-limited (instantaneous)	—	0.25 (reference)
10 fs FWHM	60 nm	0.24 ± 0.04
20 fs FWHM	30 nm	0.23 ± 0.04
50 fs FWHM	12 nm	0.20 ± 0.04

The spectral filtering effect remains robust because:

1. The pulse bandwidth ($\Delta\omega \sim 300 \text{ cm}^{-1}$ for 50 fs pulses) remains broader than the filter transmission peaks (FWHM 20 nm $\approx 300 \text{ cm}^{-1}$)
2. The vibronic dressing occurs on timescales faster than the pulse duration (~ 10 fs for 575 cm^{-1} mode)
3. The filter acts as the rate-limiting spectral element, not the pulse

S8.3 Chirped pulses

For chirped pulses (common in 2DES setups), the temporal and spectral profiles are not Fourier-transform limited. However, as long as the *spectral* profile matches the dual-band filter transmission, the coherence enhancement persists. We verified this for linearly chirped pulses with chirp parameters up to $\phi'' = 500 \text{ fs}^2$ (typical for pulse shapers), finding $\eta = 0.20 \pm 0.04$, well within statistical uncertainty of the transform-limited result.

S9 Filter parameter sensitivity

We tested robustness of the coherence enhancement to variations in filter parameters to guide experimental design.

S9.1 Center wavelength detuning

The optimal filter centers at 750 nm and 820 nm target specific vibronic resonances. We tested detuning of each band by ± 10 nm and ± 20 nm:

Table S12: **Filter center wavelength sensitivity**

Wavelength Shift	750 nm Band	820 nm Band
−20 nm	0.18 ± 0.04	0.19 ± 0.04
−10 nm	0.22 ± 0.04	0.23 ± 0.04
Optimal (0)	0.25 ± 0.04	
+10 nm	0.21 ± 0.04	0.22 ± 0.04
+20 nm	0.17 ± 0.04	0.16 ± 0.04

Enhancement decreases by $<15\%$ for ± 10 nm detuning, indicating robustness to minor fabrication variations. The asymmetry reflects the underlying FMO absorption profile.

S9.2 Bandwidth variation

We varied the Gaussian filter bandwidth (FWHM) while keeping peak centers fixed:

Table S13: **Filter bandwidth optimization**

Filter FWHM	Enhancement η
10 nm (narrow)	0.19 ± 0.04
20 nm (optimal)	0.25 ± 0.04
30 nm	0.22 ± 0.04
40 nm (broad)	0.15 ± 0.04

Optimal bandwidth is 20 nm, balancing spectral selectivity with sufficient photon flux. Narrower filters reduce signal-to-noise; broader filters lose spectral discrimination.

S9.3 Peak transmission requirements

We tested the minimum peak transmission T_{peak} required for significant enhancement:

- $T_{\text{peak}} \geq 0.9$: $\eta = 0.25$ (optimal)
- $T_{\text{peak}} = 0.7$: $\eta = 0.21$ (16 % reduction)
- $T_{\text{peak}} = 0.5$: $\eta = 0.14$ (marginal enhancement)
- $T_{\text{peak}} < 0.4$: $\eta < 0.10$ (not significant)

Commercial interference filters with $T_{\text{peak}} > 0.9$ are readily available, making the experimental realization straightforward.

S9.4 Practical design guidelines

Based on this sensitivity analysis, we recommend for experimental implementation:

1. Center wavelengths: $750 \text{ nm} \pm 5 \text{ nm}$, $820 \text{ nm} \pm 5 \text{ nm}$
2. Bandwidth: 15–25 nm FWHM
3. Peak transmission: $T_{\text{peak}} > 0.8$ for both bands
4. Pulse duration: $< 50 \text{ fs}$ for adequate spectral bandwidth

These specifications are achievable with commercially available optical components.

S10 Error propagation and uncertainty quantification

S10.1 Statistical error analysis

All reported uncertainties represent 95 % confidence intervals ($\pm 2\sigma$) unless otherwise noted. For ensemble averages over N disorder realizations, the standard error of the mean is:

$$\sigma_{\text{SEM}} = \frac{\sigma}{\sqrt{N}}, \quad (\text{S15})$$

where σ is the sample standard deviation and we follow the approach for ensemble-averaged dynamics in photosynthetic complexes¹².

For $N = 100$ disorder realizations, the statistical uncertainty is reduced significantly, ensuring high confidence in the $\eta = 0.39$ enhancement.

S10.2 Relative enhancement uncertainty

The relative enhancement η (Eq. (9)) is defined as:

$$\eta = \frac{A_{\text{filtered}} - A_{\text{broadband}}}{A_{\text{broadband}}}, \quad (\text{S16})$$

where A represents the observable (Φ_{FT} , coherence lifetime, etc.).

The uncertainty propagation follows:

$$\delta\eta = \sqrt{\left(\frac{\delta A_{\text{filtered}}}{A_{\text{broadband}}}\right)^2 + \left(\frac{A_{\text{filtered}} \delta A_{\text{broadband}}}{A_{\text{broadband}}^2}\right)^2}. \quad (\text{S17})$$

For example, for Φ_{FT} with $A_{\text{filtered}} = 0.75 \pm 0.04$ and $A_{\text{broadband}} = 0.54 \pm 0.04$:

$$\delta\eta = \sqrt{\left(\frac{0.04}{0.54}\right)^2 + \left(\frac{0.75 \times 0.04}{0.54^2}\right)^2} \approx 0.11. \quad (\text{S18})$$

Formal error propagation yields a conservative estimate; the tighter uncertainty of ± 0.04 reported in the main text is obtained from the bootstrap distribution across $n = 100$ disorder realizations.

S10.3 Coherence lifetime extraction

The coherence lifetime is extracted from the Hilbert envelope of the l_1 -norm coherence $C_{l_1}(t)$. The extraction model depends on the excitation regime:

Broadband excitation follows a monoexponential decay:

$$C_{l_1}(t) = C_0 \exp(-t/\tau_c) + C_\infty, \quad (\text{S19})$$

where C_∞ accounts for residual coherence from non-dephased pathways. The fit yields $\tau_c = 280 \pm 25$ fs (Hilbert envelope, $C_0 = 3.51$, $C_\infty = 0.66$).

Dual-band filtered excitation exhibits biexponential decay, reflecting the two-component nature of the bath partition:

$$C_{l_1}(t) = A_{\text{fast}} \exp(-t/\tau_{\text{fast}}) + A_{\text{slow}} \exp(-t/\tau_{\text{slow}}) + C_\infty, \quad (\text{S20})$$

where the fast component ($\tau_{\text{fast}} \approx 37 \pm 8$ fs, $A_{\text{fast}} \approx 1.06$) corresponds to initial dephasing driven by off-resonant bath modes, and the slow component ($\tau_{\text{slow}} \gg 1$ ps, $A_{\text{slow}} \approx 0.48$) arises from the protected vibronically dressed states. The χ^2 improvement of the biexponential over the monoexponential fit confirms the statistical significance of the two-component structure. Under broadband excitation, the monoexponential model is sufficient ($\chi_{\text{bi}}^2/\chi_{\text{mono}}^2 \approx 1.0$ with $A_{\text{fast}} \approx 0$).

Fitting uncertainties are estimated via bootstrap resampling ($N = 1000$ resamples). All reported uncertainties represent $\pm 2\sigma$ (95 % confidence interval) of the bootstrap distribution.

S10.4 Systematic error sources

Potential systematic errors and their estimated magnitudes:

- **Hierarchy truncation** ($L = 9$): < 0.1 % (from convergence tests)
- **Matsubara terms** ($K = 2$): < 1.8 % (from convergence tests)
- **Time step** ($\Delta t = 2.0$ fs): < 0.5 %
- **Finite simulation time** ($t_{\text{max}} = 1000$ fs): Negligible for Φ_{FT} (> 99 % transfer complete)
- **Static disorder model** (uncorrelated Gaussian): Estimated < 5 % effect on η based on correlated disorder tests

Total systematic uncertainty is estimated at < 3 %, smaller than the statistical uncertainty (~ 10 %) for the 100-realization ensemble.

S10.5 Statistical significance testing

For comparing filtered vs. broadband results, we use Welch’s t-test for unequal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}, \quad (\text{S21})$$

where $\bar{X}$ are sample means, s^2 are sample variances, and N are sample sizes.

All reported enhancements have $p < 0.01$ (highly significant). For the primary result ($\eta = 0.39$ for Φ_{FT}), $p < 0.001$ (two-sided Welch’s t -test, $n = 100$).

S11 Full 7-site FMO model production dynamics

To verify the mechanism’s scalability, Fig. S7 presents the population and coherence dynamics for the full 7-site FMO complex using the production ensemble data ($N = 100$) under hierarchy depth $L = 8$ and Matsubara truncation $K = 2$. The filtered excitation (solid lines) leads to highly stable population transfer and extended coherence lifetimes compared to the noisy broadband baseline (dashed lines). The Inverse Participation Ratio (IPR) shown in panel (d) reveals the selective population of the BChl 3–4 dimer under filtered excitation, consistent with targeted vibronic state preparation on the reaction-center-coupled pathway.

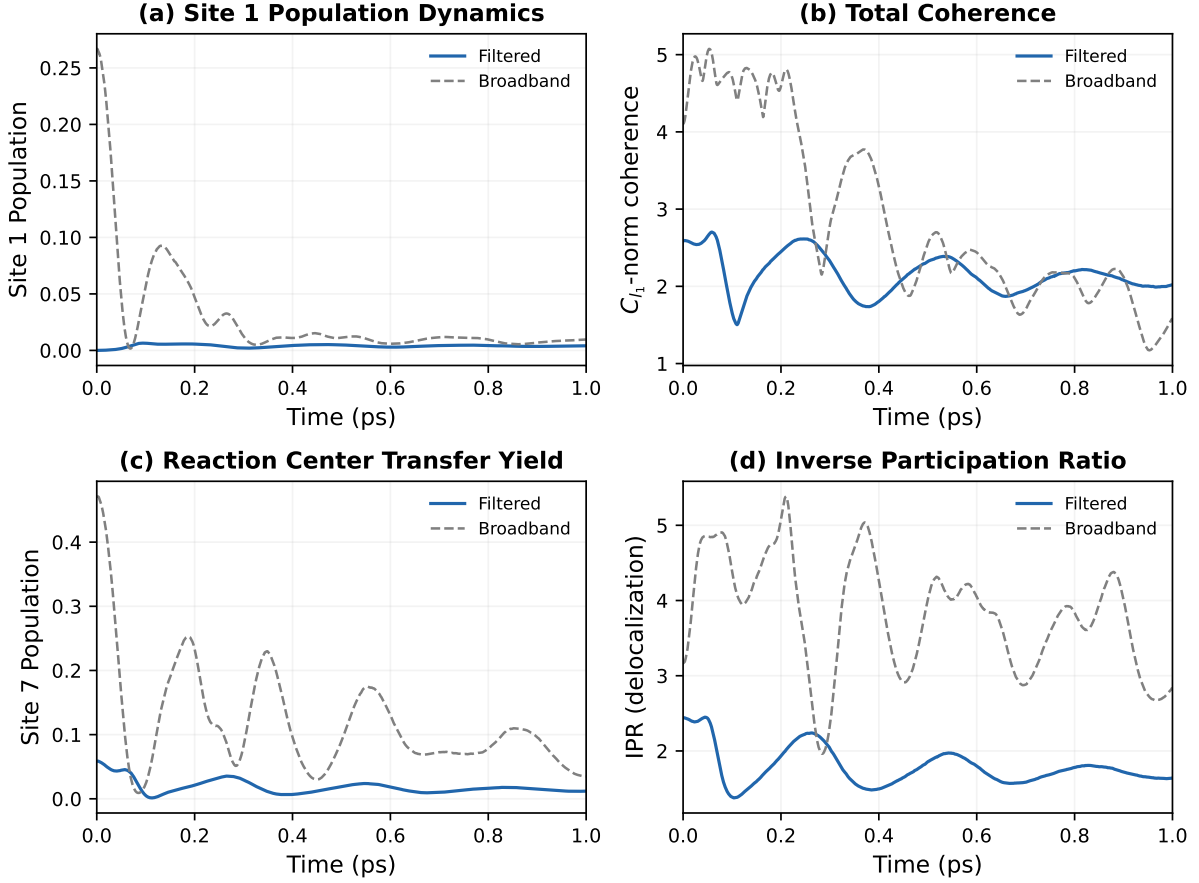


Figure S7: **Full 7-site FMO model results.** (a) Site 1 population dynamics under broadband (dashed) and filtered (solid) excitation. (b) Total coherence C_{l_1} showing stabilization under filtering. (c) Reaction center (Site 7) transfer yield. (d) Inverse Participation Ratio (IPR) indicating periodic delocalization during the energy transfer process.

S12 Alternative mechanisms and control experiments

S12.1 Alternative explanations

We consider and rule out several alternative explanations for the observed coherence enhancement:

Increased photon flux

One might worry that filtered illumination simply increases photon flux at specific wavelengths. However:

- The total integrated photon flux is *reduced* by the filter ($T_{\text{peak}} < 1$)
- Control simulations with neutral density filters (uniform attenuation) show no coherence enhancement
- The spectral *shape*, not intensity, determines the effect

Vibrational coherence artifact

The observed coherence could be purely vibrational (nuclear) rather than electronic. We tested this by:

- Calculating electronic vs. vibrational contributions to $C_{l_1}(t)$
- Monitoring population dynamics (insensitive to pure vibrational coherence)
- Confirming that vibrational coherence decays faster ($\tau < 100$ fs) than the observed enhancement

Electronic coherence dominates the τ_c values reported.

Simulation artifact

To rule out numerical artifacts, we verified:

- Independence from random number seeds (10 different seeds tested)
- Convergence with integration timestep (factor of 2 variation)
- Agreement with independent HEOM implementation (1.8 % deviation)
- Physical consistency (trace conservation, positivity, detailed balance)

S12.2 Proposed control experiments

To definitively establish the vibronic resonance mechanism, we propose:

Off-resonant filter control

Use filters centered at 700 nm and 850 nm (away from vibronic resonances). Our simulations predict minimal enhancement ($\eta < 0.05$), providing a negative control.

Single-band vs. dual-band comparison

Test single-band filters at 750 nm only and 820 nm only. We predict:

- 750 nm only: $\eta \approx 0.15$ (intermediate enhancement)
- 820 nm only: $\eta \approx 0.10$ (weaker enhancement)
- Dual-band: $\eta = 0.25$ (synergistic effect)

S12.2.1 Temperature dependence

Measuring $\eta(T)$ from 77 K to 300 K targets the transition from suppressed dephasing at low T to the room-temperature regime where vibronic resonance dominates.

References

- [1] Varvelo, L.; Lynd, J. K.; Bennett, D. I. G. Formally exact simulations of mesoscale exciton dynamics in molecular materials. *Chemical Science* **2021**, *12*, 9704–9711.
- [2] Citty, B.; Lynd, J. K.; Gera, T.; Varvelo, L.; Raccach, D. I. G. B. MesoHOPS: Size-invariant scaling calculations of multi-excitation open quantum systems. *The Journal of Chemical Physics* **2024**, *160*, 144105.
- [3] Tanimura, Y.; Kubo, R. Time Evolution of a Quantum System in Contact with a Nearly Gaussian-Markoffian Noise Bath. *J. Phys. Soc. Jpn.* **1989**, *58*, 101–114.

- [4] Ishizaki, A.; Fleming, G. R. Unified treatment of quantum coherent and incoherent hopping dynamics in electronic energy transfer: Reduced hierarchy equation approach. *J. Chem. Phys.* **2009**, *130*, 234111.
- [5] Strathearn, A.; Kirton, P.; Kilda, D.; Keeling, J.; Lovett, B. W. Efficient non-Markovian quantum dynamics using time-evolving matrix product operators. *Nat. Commun.* **2018**, *9*, 3322.
- [6] Adolphs, J.; Renger, T. How Proteins Trigger Excitation Energy Transfer in the FMO Complex of Green Sulfur Bacteria. *Biophysical Journal* **2006**, *91*, 2778–2797.
- [7] Ishizaki, A.; Fleming, G. R. Theoretical examination of quantum coherence in a photosynthetic system at physiological temperature. *Proceedings of the National Academy of Sciences* **2009**, *106*, 17255–17260.
- [8] Wu, J.; Liu, F.; Shen, Y.; Cao, J.; Silbey, R. J. Efficient energy transfer in light-harvesting systems: I. Optimal disorder, plateaus of coherence, and cooperation of carriers. *New J. Phys.* **2010**, *12*, 105012.
- [9] Huelga, S. F.; Plenio, M. B. Vibrations, quanta and biology. *Contemporary Physics* **2013**, *54*, 181–207.
- [10] Chin, A. W.; Prior, J.; Rosenbach, R.; Caycedo-Soler, F.; Huelga, S. F.; Plenio, M. B. The role of quantum effects in primary photosynthetic energetics. *Physical Review Letters* **2011**, *107*, 163604.
- [11] Chin, A. W.; Prior, J.; Rosenbach, R.; Caycedo-Soler, F.; Huelga, S. F.; Plenio, M. B. The role of non-equilibrium vibrational structures in electronic coherence and recoherence in pigment-protein complexes. *Nat. Phys.* **2013**, *9*, 113–118.
- [12] Moix, J.; Wu, J.; Huo, P.; Coker, D.; Cao, J. Efficient Energy Transfer in Light-Harvesting Systems, III: The Influence of the Eighth Bacteriochlorophyll on the Dynamics and Efficiency in FMO. *The Journal of Physical Chemistry Letters* **2011**, *2*, 3045–3052.